

1 Signals and Systems

Signals A signal is a function of one or more *independent* variables.

- Continuous-time signals: $x(t)$, where t is a continuous variable.
- Discrete-time signals: $x[n]$, where n is a discrete variable.

Systems A system is a process that transforms signals.

2 Energy and Average Power

For a signal $x(t)$ or $x[n]$ (usually takes on *complex values*), the energy and average power are defined as:

2.1 Discrete-Time

Definition (Energy).

Over a time interval $[n_1, n_2]$:

$$\sum_{n=n_1}^{n_2} |x[n]|^2$$

Over an infinite time interval:

$$E_{\infty} := \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Definition (Average Power).

Over a time interval $[n_1, n_2]$:

$$\frac{1}{n_2 - n_1 + 1} \sum_{n=n_1}^{n_2} |x[n]|^2$$

Over an infinite time interval:

$$P_{\infty} := \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=-N}^N |x[n]|^2$$

2.2 Continuous-Time

Definition (Energy).

Over a time interval $[t_1, t_2]$:

$$\int_{t_1}^{t_2} |x(t)|^2 dt$$

Over an infinite time interval:

$$E_{\infty} := \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Definition (Average Power).

Over a time interval $[t_1, t_2]$:

$$\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

Over an infinite time interval:

$$P_{\infty} := \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

2.3 Cases of Energy and Average Power

	E_{∞}	P_{∞}	Remark
Case 1	0	0	iff the signal is always zero.
Case 2	finite	0	
Case 3	∞	finite	
Case 4	∞	∞	

3 Transformations of the Independent Variable

- Time Shift
- Time Reversal
- Time Scaling

4 Periodic and Aperiodic Signals

If a signal $x(t)$ or $x[n]$ is periodic with period T or N , where T or N is positive, then

$$x(t + T) = x(t) \quad \text{or} \quad x[n + N] = x[n]$$

Fundamental Period T_0 or N_0 : the smallest positive value of T or N .

5 Even and Odd Signals

Definition (Even-Odd Decomposition).

$$\mathcal{E}v\{x[n]\} := \frac{1}{2} [x[n] + x[-n]]$$

$$\mathcal{O}d\{x[n]\} := \frac{1}{2} [x[n] - x[-n]]$$

$$\mathcal{E}v\{x(t)\} := \frac{1}{2} [x(t) + x(-t)]$$

$$\mathcal{O}d\{x(t)\} := \frac{1}{2} [x(t) - x(-t)]$$

Then we have:

$$x[n] = \mathcal{E}v\{x[n]\} + \mathcal{O}d\{x[n]\}$$

$$x(t) = \mathcal{E}v\{x(t)\} + \mathcal{O}d\{x(t)\}$$

6 Exponential and Sinusoidal Signals

6.1 General Complex Exponential Signals

Let $C = |C|e^{j\theta}$ and $a = r + j\omega_0$.

$$x(t) = Ce^{at} = |C|e^{rt} \left[\cos(\omega_0 t + \theta) + j \sin(\omega_0 t + \theta) \right]$$

Let $C = |C|e^{j\theta}$ and $\alpha = |\alpha|e^{j\omega_0}$.

$$x[n] = C\alpha^n = |C||\alpha|^n \left[\cos(\omega_0 n + \theta) + j \sin(\omega_0 n + \theta) \right]$$

6.2 periodicity

6.2.1 Continuous-Time

- A complex exponential signal $x(t) = Ce^{at}$, where $a = r + j\omega_0$, is periodic **iff** $r = 0$.
- A sinusoidal signal $x(t) = A \sin(\omega_0 t + \phi)$ is **always** periodic.
- When $\omega_0 \neq 0$, the *fundamental period* for both of them is $T_0 = \frac{2\pi}{|\omega_0|}$.

Remark. Actually, the sinusoidal signal is just a special case of the complex exponential signal, where $r = 0$, thus it must be periodic.

6.2.2 Discrete-Time

- A complex exponential signal $x[n] = C\alpha^n$, where $\alpha = |\alpha|e^{j\omega_0}$, is periodic **iff** $|\alpha| = 1$ and $|\omega_0| = \frac{2\pi m}{N}$ for some $N, m \in \mathbb{N}$ (i.e., ω_0 must be a rational multiple of π).
- A sinusoidal signal $x[n] = A \sin(\omega_0 n + \phi)$ is periodic **iff** $|\omega_0| = \frac{2\pi m}{N}$ for some $N, m \in \mathbb{N}$ (i.e., ω_0 must be a rational multiple of π).
- When $\omega_0 \neq 0$, the *fundamental period* for both of them is $N_0 = \frac{2\pi m}{|\omega_0|}$, where m is the least integer such that $N_0 \in \mathbb{Z}_+$.

Remark. The sinusoidal signal is just a special case of the complex exponential signal, where $|\alpha| = 1$.

7 Unit Step and Unit Impulse Functions

7.1 Discrete-Time

Definition (Unit Step Function).

$$u[n] := \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$u[n]$ is the *running sum* of $\delta[n]$, i.e.,

$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

Definition (Unit Impulse Function).

$$\delta[n] := \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

$\delta[n]$ is the *first difference* of $u[n]$, i.e.,

$$\delta[n] = u[n] - u[n-1]$$

7.2 Continuous-Time

First we define:

$$u_{\Delta}(t) := \begin{cases} 0, & t < 0 \\ \frac{t}{\Delta}, & 0 \leq t \leq \Delta \\ 1, & t > \Delta \end{cases}$$

$$\delta_{\Delta}(t) := \frac{du_{\Delta}(t)}{dt}$$

Then we can define:

Definition (Unit Step Function).

$$u(t) := \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

$u(t)$ is the *running integral* of $\delta(t)$, i.e.,

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Definition (Unit Impulse Function).

$$\delta(t) := \lim_{\Delta \rightarrow 0^+} \delta_{\Delta}(t)$$

$\delta(t)$ is the *first derivative* of $u(t)$, i.e.,

$$\delta(t) = \frac{du(t)}{dt}$$

7.3 Properties

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0) \quad \int_{-\infty}^t f(\tau)\delta(\tau) d\tau = f(0)u(t)$$

Note that the Fundamental Theorems of Calculus do not hold as $f(\tau)\delta(\tau)$ is not continuous.

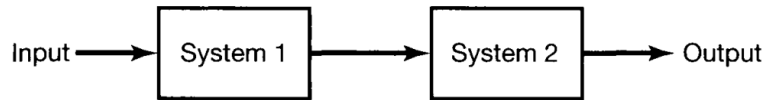
$$\begin{aligned} \int_a^b f(\tau)\delta(\tau) d\tau &= f(0) [u(-a)u(b) - u(a)u(-b)] \\ \int_a^b f(\tau)u(\tau) d\tau &= \int_{u(a)a}^{u(b)b} f(\tau) d\tau \end{aligned}$$

8 Interconnections of Systems

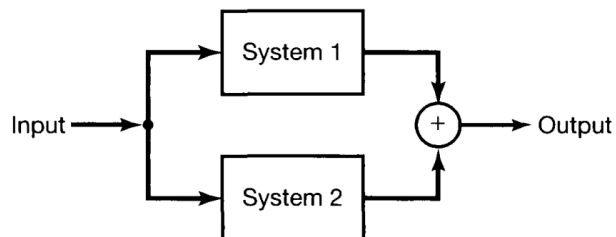
We can use *block diagrams* to represent the interconnections of systems.

8.1 Series (Cascade) Interconnection

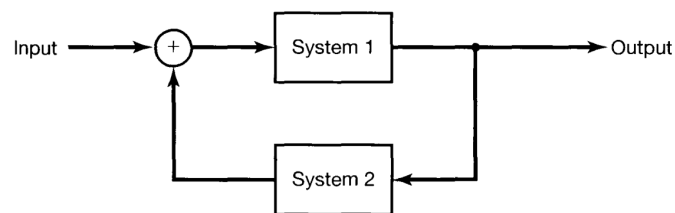
A *series* or *cascade interconnection* of two systems is shown in the following block diagram:



8.2 Parallel Interconnection



8.3 Feedback Interconnection



9 Basic System Properties

9.1 Memory

Definition (Memoryless).

A system is said to be *memoryless* if its output for each value of the independent variable at a given time is dependent **only on the input at that same time**.

9.2 Invertibility

Definition (Invertible).

A system is said to be *invertible* if distinct inputs lead to distinct outputs.

9.3 Causality

Definition (Causal).

A system is *causal* if the output at any time depends only on values of the input at the present time and in the past.

Such a system is often referred to as being *nonanticipative*, as the system output does not anticipate future values of the input.

9.4 Stability

Definition (Stable).

A system is *stable* if **bounded input** gives **bounded output** (BIBO).

9.5 Time Invariance

Definition (Time-Invariant).

A system is *time-invariant* if a time shift in the input only causes a time shift in the output. Otherwise, it is *time-variant*.

A system S is *time-invariant* iff it satisfies:

$$x(t) \xrightarrow{S} y(t) \implies x(t + \tau) \xrightarrow{S} y(t + \tau)$$

or

$$x[n] \xrightarrow{S} y[n] \implies x[n + n_0] \xrightarrow{S} y[n + n_0]$$

9.6 Linearity

Definition (Linear).

A system S is *linear* if it satisfies the following two properties:

- *Additivity*: $x_1(t) + x_2(t) \xrightarrow{S} y_1(t) + y_2(t)$
- *Scaling or Homogeneity*: $a \cdot x(t) \xrightarrow{S} \cdot y(t)$

If linear, then **zero input** gives **zero output** (ZIZO).

9.7 Linear Time-Invariant (LTI) Systems

Systems that are both linear and time-invariant.